TOOLS CHEAT SHEET

ADAMBOTS Team 245

WRITE	SAVE	SHARE	BUILD	TEST	DEPLOY	DRIVE
VS Code	Git commit	GitHub	Gradle	Simulation	SystemCore	Driver Station

1 VS Code shortcuts

DO THIS	WINDOWS	MAC
Find any command	`Ctrl+Shift+P`	`Cmd+Shift+P`
Fix it (light bulb)	`Ctrl+.`	`Cmd+.`
Go to Definition	`F12` / `Ctrl`+click	`F12` / `Cmd`+click
Jump back	`Alt+←`	`Ctrl+-`
Rename everywhere	`F2`	`F2`
Format Document	`Shift+Alt+F`	`Shift+Option+F`
Search all files	`Ctrl+Shift+F`	`Cmd+Shift+F`
Problems panel	`Ctrl+Shift+M`	`Cmd+Shift+M`
Source Control	`Ctrl+Shift+G`	`Cmd+Shift+G`
Save	`Ctrl+S`	`Cmd+S`

Always use **WPILib VS Code**: look for the WPILib logo, top right.

2 Build, simulate, deploy

Build Robot Code	WPILib menu. Read the last line: `BUILD SUCCESSFUL`. First build must be online.
Simulate Robot Code	Pick Sim GUI. Choose Teleoperated in Robot State.
Deploy Robot Code	`Shift+F5`. Finds robot 245. RioLog shows the console.
Weird build?	WPILib menu → **Run Gradle Clean**, build again.

> **Never power off the robot during a deploy.** It can corrupt SystemCore.

3 Project map

`src/main/java/first/robot/`	all our code: edit every day
`Constants.java`	every fixed number, named `kSomething`
`Robot.java`	builds the subsystems, runs the scheduler
`subsystems/`	one class per mechanism
`opmode/`	the OpModes the Driver Station lists
`vendordeps/`	vendor libraries (Manage Vendor Libraries)
`.wpilib/`	our team number lives here
`build/`, `build.gradle`	never edit

4 Our Git workflow (all in VS Code)

1. **Pull:** on `main`, click **Sync** (bottom-left).
2. **Branch:** click the branch name → Create new branch → `yourname-topic`
3. **Commit:** Source Control → **+** to stage → message → **Commit**. Small steps.
4. **Push: Publish Branch** the first time, then **Sync Changes**.
5. **Pull request** on github.com/Adambots-245 → a mentor reviews and merges.

> Nobody pushes straight to `main`. Messages finish the sentence "This commit will...": *Stop intake when beam break sees a piece.*

5 Words

repository	project plus its whole history	**clone**	download a repo
commit	save point	**branch**	your own line of work
push	send commits up	**pull**	bring commits down
pull request	"review and merge my branch"	**merge**	combine branches

6 Merge conflicts

```
<<<<<<< HEAD              current: already here
kIntakeSpeed = 0.8;
=======
kIntakeSpeed = 0.6;           incoming: being merged
>>>>>>> ben-intake-speed
```

Talk to your teammate. Click **Accept Current**, **Incoming** or **Both**, or keep the right line and delete all three marker lines. Then save, stage, commit, Sync. Avoid them: pull often, small branches, one person per file.

7 Driver Station and safety

E-Stop	`Space` disables at once. Reset: hold `Esc` + `I` for 1 second.
A-Stop	orange button: ends autonomous early
Battery	our team's rule: **12.0 V or more** is match-ready
Modes	Tele · Auto · Util · Match
Dashboards	Elastic (live) · AdvantageScope (graphs, logs)

1 Look	2 Call	3 E-Stop
everyone clear, glasses on	"Enabling!" … "Clear!"	hand on the spacebar